

Neural network experiment defense

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone limitations and revision

Without opening the old lesson, explain one key idea from Capstone limitations and revision. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Capstone hypothesis and baseline

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset shift in neural models

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Defend the result for Neural network experiment defense. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Neural network experiment defense. Proposed explanation: “A polished demonstration is an evaluation”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Defend a new record independently

A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The written rule selects lower MSE. After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction. Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Build the evidence inventory

Name eight items needed to check the capstone from question through final result.

My reasoning and evidence

6. W2 Defend the result

State the validation selection and frozen final comparison using their correct metrics and scopes.

My reasoning and evidence

7. W3 Challenge an unsupported claim

A presenter claims the experiment proves neural networks beat fitted lines and use less energy. What evidence is missing?

My reasoning and evidence

8. W4 Defend a new record independently

A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The written rule selects lower MSE. After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction. Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

My reasoning and evidence

9. W5 Record the next investigation

Propose a fair next comparison with a fitted signed line. State how to preserve previous evidence and reserve new evaluation.

My reasoning and evidence

10. Project checkpoint and reflection

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final evidence pack: A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The written rule selects lower MSE. After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction. Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Neural network experiment defense

Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

Inputs are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, $\max(0,z)$. This is a demonstration, not a probability model.

One toy ReLU neuron: not a probability

Input 2 x weight 0.5 = 1

Input 1 x weight -1 = -1

Add the products and bias 0.2: total = 0.2

ReLU: use $\max(0, \text{total})$ -> output 0.2

Neural network experiment defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Multiply inputs

$2 \times 0.5 = 1$ and $1 \times (-1) = -1$.



2. Add weighted terms

Pre-activation z is $1 - 1 + 0.2 = 0.2$.



3. Apply ReLU

$\max(0, 0.2) = 0.2$; this activation is not a confidence percentage.

CHANGE ONE THING

Change the second input from 1 to 2.

Explain the changed result using the matching lesson.

Calculate both weighted terms, z and ReLU output. Predict the changed second input and explain why 0.2 cannot be called 20% accuracy.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

